$$J(\theta) = \frac{1}{2}\sum_{i=1}^{n}\left(h_\theta(x^{(i)}) - (y^{(i)})\right)^2$$

going down

Gradient

Nabla

$$\theta_0 := \theta_0 \boxed{-\alpha} \boxed{\frac{\partial J}{\partial \theta_0}} \Rightarrow \theta_0 - \alpha \boxed{\nabla_0 J_0}$$

Steps (Learning rate)

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_n \end{bmatrix}_{(n+1)\times 1}$$

$\alpha \Rightarrow$ Hyperparameter

Hyperparameter tuning

$\alpha = 0.1,\ 0.01,\ 0.001$

Gradient descent Algorithm / Regular gradient descent algorithm / Batch " " "

j = dimension (0 to d)

Repeat until convergence.

$$\{\ \theta_j := \theta_j - \alpha \sum_{i=1}^{n} (h_\theta(x^{(i)}) - y^{(i)}) * x_j^{(i)}\ \}$$

$\Rightarrow$ For every (j)

$$\theta_0 = \theta_0 - \alpha \sum_{i=1}^{n} [h_\theta(x^{(i)} - y^{(i)})\, x_0^{(i)}$$

$$\theta_1 = \theta_1 - \alpha \sum_{i=1}^{n} (h_\theta(x^{(i)} - y^{(i)})\, x_1^{(i)}$$

$$\theta_2 = \theta_2 - \alpha \sum_{i=1}^{n} (h_\theta(x^{(i)} - y^{(i)})\, x_2^{(i)}$$

$\theta_0 1 + \theta_1 1 + \theta_2 2 = 3$

x_0	x_1	x_2	y
1	1	2	3
1	2	1	4
1	3	3	5

$\alpha = 0.1$

$n = 3$

$d = 2$

$x \in \mathbb{R}^{d+1}$

for a single sample

$$\frac{\partial J}{\partial \theta_j} = \frac{\partial}{\partial \theta_j}\left[\frac{1}{2}\left(h_\theta(x) - y\right)^2\right]$$

$$= \frac{\partial}{\partial \theta_j}\left[\frac{1}{2}\left((\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_d x_d) - y\right)^2\right]$$

$$\frac{\partial J}{\partial \theta_0} = \frac{\cancel{2}}{\partial \theta_0}\left[\frac{1}{\cancel{2}} \cdot \cancel{2}(\theta_0 x_0 + \cdots + \theta_d x_d) - y\right] \cdot x_0$$

$$\frac{\partial J}{\partial \theta_1} = \left[\frac{1}{\cancel{2}} \cdot \cancel{2}(\theta_0 x_0 + \cdots + \theta_d x_d) - y\right] \cdot x_1$$

0.4 $\quad 0.4 + 0.81 \times 1 + 0.83 \times 2$

0.81

1.66

Apply gradient descent

Iteration 1

$\theta_0 = \theta_1 = \theta_2 = 0$

update $\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$

Iteration 2

$\theta_0 =$; $\theta_1 =$; $\theta_2 =$

update $\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$

$$\theta_0 = \theta_0 - \alpha \sum_{i=1}^{n} (h_\theta(x^{(i)}) - y^{(i)})^2 \cdot x_0^{(i)}$$

$$= \theta_0 - 0.1 (h_\theta x^1 - y^1)^2 \cdot x_0^1$$

$$= \theta_0 - 0.1 \left[(h_\theta x^1 - y^1)^2 + (h_\theta x^2 - y^2)^2 + (h_\theta x^3 - y^3)^2 \right] \cdot x_0^1$$

$$= \theta_0 - 0.1 \left[(h_\theta x^1 - y^1)^2 \cdot x_0^1 + (h_\theta x^2 - y^2)^2 \cdot x_0^2 + (h_\theta x^3 - y^3)^2 \cdot x_0^3 \right]$$

$$\theta_0 = \theta_1 = \theta_2 = 0$$

$$= 0 - 0.1 \left[(0-3)^2 \cdot 1 + (0-4)^2 \cdot 1 + (0-5)^2 \cdot 1 \right]$$

$$= 0 - 0.1 \left[\overset{-3\ -4\ -5}{9 + 16 + 25} \right] = 0 - 0.1 [-12] = 1.2$$

$$\theta_0 = 0 - 5 \Rightarrow -5$$

$$\theta_1 = \theta_1 - 0.1 \left[(h_\theta x^1 - y^1)^2 + (h_\theta x^2 - y^2)^2 + (h_\theta x^3 - y^3)^2 \right] \cdot x_1$$

$$= 0 - 0.1 \left[(0-3)1 + (0-4)2 + (0-5)3 \right]$$

$$= 0 - 0.1 [-3 - 8 - 15]$$

$$= 0 - 0.1 [-26]$$

$$= 0 + 2.6 \Rightarrow 2.6$$

$$\theta_2 = 0 - 0.1 \left[(0-3) \cdot 2 + (0-4) 1 + (0-5) 3 \right]$$

$$= 0 - 0.1 [-6 - 4 - 15]$$

$$= 0 - 0.1 [-25]$$

$$= 0 + 2.5$$

$$\theta_2 = 2.5$$

$$\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1.2 \\ 2.6 \\ 2.5 \end{bmatrix}$$

$$h_\theta(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

$$= (1.2)(1) + (2.6)(1) + (2.5)(2)$$

$$= 1.2 + 2.6 + 5.0$$

$$= 8.8$$

S.No	x_0	x_1	x_2	y	$\hat{y}$ $h(\theta)$
1	1	1	2	3	8.8
2	1	2	1	4	8.9
3	1	3	3	5	16.5

$$\theta_0 = 1.2 - 0.1 \left[(1.2 - 3)1 + (2.6 - 4)1 + (2.5 - 5)3 \right]$$

$$\theta_0 = 1.2 - 0.1 [-1.8 - 1.4 - 2.5]$$

$$= 1.2 - 0.1 [-5.7]$$

$$= 1.2 + 0.57$$

$$+ 2 x_2 + 3 C_2 + 3 = 2$$

$$x_0 + x_1 - x_2 \; 3 x$$

$(1.2)(1) + (2.6)$

$\theta_0 = -1.02$

$\theta_1 = -2.41$

$\theta_2 = -2.6$

$\theta_0 = 1.2 - 0.1\left[(8.8^{(1)} - 3)^1 + (8.8^{(2)} - 4)^1 + (8.8^{(3)} - 5)^1\right]$

$17.6 - 4 \qquad 26.4$

$= 1.2 - 0.1(5.8 + 4.8 + 3.8)$

$= 1.2 - 0.1(14.4)$

$= 1.2 - 1.44$

$= 1.2 - 0.1(5.8 + 13.6 + 21.4)$

$= 1.2 - 4.08$

$= -2.88$

$\theta_0 = \theta_0 - 0.1\left[h_\theta x^1 - y^1_{x_0} + h_\theta x^2 - y^2_{x_0} + h_\theta x^3 - y^3_{x_0}\right]$

$= 1.2 - 0.1\left[8.8(1) - 3 + 8.8(2) - 4 + 8.8(3) - 5\right]$

$(1.2)(1) + (2.6)(2) + (2.5)(1)$

$\Downarrow$

$1.2 + 5.2 + 2.5 \Rightarrow 8.9$

$1.2 + 7.8 + 7.5 \Rightarrow 16.5$

$1 \times 1.2 + 1 \times 2.6 + 2 \times 2.5 =$

$$\begin{array}{r} 1.2 \\ 2.6 \\ 2.5 \\ \hline 7.2 \end{array}$$

$(h_\theta x^1)$